

vasic-digital tmux — v1.0.45

Release Notes

Revision: 3 **Last modified:** 2026-09-02T06:32:13Z **Version:** 1.0.45 / versionCode 46 **Released:** NOT YET — tag `tmux-1.0.45` is not cut. Intended 2026-09-01. **Hosts validated:** the full `verify.sh` sweep COMPLETED GREEN on this tree (PASS=79 FAIL=0 SKIP=11, 2026-09-02T06:09Z) — see "Validation" below for the result, the enumerated topology skips, and its honest boundary. §11.4.185 manual QA has NOT been performed and is not claimed (§11.4.6).

Summary

v1.0.45 is a **workable-items tracker-integrity** release. The tmux product surface is UNCHANGED from v1.0.44 — the same hardened `tmux 3.7b` binary, the same `tmx` wrapper, the same submodule pin (`e802909de06012a4df6209d55e86487c56223163`). Every change is in the §11.4.93 SQLite single-source-of-truth engine (`cmd/workable-items/`), the release gate (`scripts/verify.sh`), and the tracker corpus itself.

The cycle's defects share one shape: **a parser, a check, or a gate returning a confident quiet zero while enforcing nothing**. A heading regex that made a whole class of tracker block invisible to the SSoT; an identity audit that discarded a second ownership claim before it could be reported; two shipped governance gates that no runner ever invoked; and a test assertion that could not distinguish a skipped audit from a clean one. Fixing the first exposed six more defects in the sync engine that its narrowness had been masking.

This release is not finished. The independent review returned NO-GO and has not yet iterated to a clean GO (§11.4.134), four reviewer-authored mutations survived the author's test suite, and several proven root causes have fixes that are OWED rather than done. All of it is enumerated under "Known-open" below rather than implied complete.

Fixed

Parser / SSoT

- **Space-form headings were silently dropped from the SSoT.**
parser.go's headingRE required a period after the block ordinal (`^###\s+([A-Z])(\d+)\.\s+(.*)$`), so `### G5 NAME-001 – ...` never parsed. Measured consequence: the block's body was absorbed into the PRECEDING item's raw_body, and its own row sat in the DB with a sentinel identity (category Z / code_ordinal 0), so nothing could locate it in the markdown. The corpus carries both forms — 23 of the Fixed.md block headings were space (measured at tag `tmux-1.0.44`; the corpus repair in this same release adds more) form — and the reconciler's own matcher `blockCodeRE` had ALWAYS accepted both, so the narrow parser was an internal inconsistency between two components of one tool, not a deliberate contract. Widened to `^###\s+([A-Z])(\d+)(?:\.\s+|\s+)(\S.*)$`.
- **Both regexes were TIGHTENED in the same change and are now changed together by construction.** `blockCodeRE` went from `^###\s+([A-Z])(\d+)[.\s]` to `^###\s+([A-Z])(\d+)(?:\.\s+|\s+)\S`, so the two agree on their accepted set. The separator is REQUIRED and must be followed by real title text, which correctly refuses `### A50X`, `### A1.x`, `### A12.5` (a decimal token would have yielded ordinal 12, title "5 ..."), `### A1.` and `### A1` (empty title — its `heading_hash` would be computed over ""). Blast radius was measured BEFORE any sync: 90 loose-accepted corpus headings scanned, 0 dropped, control-needle proven (§11.4.201(7)(b)).
- The contract comment above `headingRE` is declared the §11.4.245 **specified oracle** for `heading_forms_test.go`: that test's accept/refuse table is derived from the prose contract, not from the regex, so a change to one that is not a change to the other FAILs.

The six defects the widening exposed (each fixed at root cause)

#	Defect	Fix
1	UpsertItem decided INSERT-vs-UPDATE on heading_hash alone, so a newly-parseable block died on UNIQUE constraint failed:	resolve an explicit TMX-ID to its existing row, UPDATE it, and REFRESH the hash so the next sync binds on the hash instead of repeating the miss

#	Defect	Fix
	items.atm_id (reproduced live for TMX-072)	
2	Two blocks for one item silently COLLAPSED into one row whose survivor depended on read order	fail-closed guard over the UNION of ExplicitATM and HeadingHash across every parsed block in BOTH files — it REFUSES rather than picking a winner
3	That guard persisted document_sources BEFORE running, so a REFUSED sync still mutated the git-tracked SSoT	raw text held in locals; both blobs persisted only after the guard passes (§11.4.252)
4	An identity rebind replaced a row's category / ordinal / title / body with NO history row and NO counter — validate reported 0 findings because the result was self-consistent	prior identity captured, item_history row written, counted in SyncResult.RebindDetails, printed by the CLI (§11.4.226)
5	itemContentEqual compared Category but not CodeOrdinal / HeadingHash, so an ordinal-only renumber took the "unchanged" path and the row kept a stale ordinal + hash PERMANENTLY	both fields compared (identity IS content); rows self-heal from either drift
6	lookupByHeadingHash matched on error TEXT and returned ("", nil) for real DB errors — a failed query read as a hash MISS, which mints a duplicate row	errors.Is(err, sql.ErrNoRows); real errors propagate

add / db-to-md seams

- **TMX-094's fix was structurally suppressed on the live corpus — an add-created row still rendered into NO tracker (found during release verification, 2026-09-01).** sourcelessItems derived a block code from an item's stored code_ordinal, but every workable-items add row carries the SENTINEL code_ordinal = 0, rendering as code A0 — and Fixed.md:2585 holds a genuine

A0. Initial migration from ATMOSphere project ... block declaring no ****TMX-ID:****. codeOwners["A0"][""] therefore answered for EVERY add-created row in category A, suppressing exactly the rows TMX-094 exists to render. The codebase's own rule at validate_identity.go:163 ("ordinal sentinel — asserts no block claim") was contradicted. Measured: TMX-099/TMX-100 sat in the DB with current_location = Issues rendering into neither tracker (sourceless count=0 against the real corpus). Fixed by skipping the stored-ordinal check when CodeOrdinal <= 0; the atmOrdinal fallback is untouched. Post-fix: both render as ### A99. / ### A100., issues_parsed 10 → 12, round-trip byte-identical, validate 0 findings. Guarded by a §11.4.115 polarity test (RED_MODE=1 reproduces count=0) plus a §11.4.201(1) false-positive guard, proven load-bearing by a paired §1.1 mutation. The original per-signal tests missed it because their synthetic fixtures contained no <CAT>0 block — a fixture-realism gap (§11.4.245).

- **TMX-093 — add hashed the ATM id where the parser hashes the BLOCK CODE.** add.go passed atm into computeHeadingHash, while parser.go builds code := category + ordinal (parser.go:186) before hashing it (parser.go:201). An add-created row could not bind to its own block by hash and survived only via the ExplicitATM rebind fallback — which REPLACES the row and refreshes the hash, **masking** the split behind something that looked like normal operation. Fixed by hashing the block code the writer actually emits (category + atmOrdinal(atm), mirroring sync_db_to_md.go:195-201).
- **TMX-095 (NEW) — add --category accepted values rendering a heading the parser can never read.** Measured 2026-09-01: --category=AB renders ### AB1., --category=1 renders ### 11.; neither matches headingRE. The row lands in the SSoT and the writer emits its block, but the parser cannot see it — the same invisible-block class, reached through the add seam. AddItem now refuses anything that is not exactly one letter A-Z, with an actionable message; A, Z, lowercase a, and the empty default MUST still be accepted (§11.4.201(1) — a guard that refuses the real corpus is a defect, not a safe default). Surfaced as an honest out-of-scope gap by the stream fixing TMX-093 and FILED rather than narrated (§11.4.197).
- **TMX-094 — db-to-md never rendered add-created rows.** reconcileLocations only MOVES blocks that already exist in a blob; it never APPENDS one for an item present in items but in NEITHER blob. Since add writes an items row and nothing to

document_sources, an add-created row was a DB row no tracker document ever showed — a silent loss of visibility at the requirements-intake layer (§11.4.202 / §11.4.197). Fixed with sourcelessItems() + appendRenderedItems(), gated on THREE independent signals because a false positive here DUPLICATES a block into the tracker, which is strictly worse than the omission it repairs: (a) items.raw_body empty; (b) no TMX-ID line anywhere in either blob; (c) no plausibly-own ### <CAT><N> block — a bare code match is NOT sufficient, because heading codes are not unique (measured: 93 code headings, 85 distinct — 8 codes appear twice), and the ordinal is checked BOTH as the stored CodeOrdinal and as writeItemBlock's atmOrdinal fallback because the two legitimately disagree for legacy rows. Signals (b) and (c) are each individually insufficient: the id-line signal alone flags 62 of 95 items, **61 of them false**. Measured on the live corpus: **0** rows have an empty raw_body, so the pass appends nothing today and the live regeneration stays byte-identical.

Identity audit

- **The §11.4.54 audit DISCARDED a second ownership assertion in silence.** markdownBlockOwners was first-declarer-wins: Fixed.md carried 25 blocks declaring a TMX-ID while the owner map held 24 keys — ### A52 headed two blocks and one claim vanished, so the audit could not report a collision it had already thrown away. Duplicates are now RETURNED and raised as findings naming every claimant. Cross-FILE reuse of a code is deliberately NOT a duplicate — identity is (location, category, ordinal), and the live corpus legitimately reuses 8 codes across the two trackers (§11.4.201(1)).
- **A code-less ### heading did not reset the id-scan cursor**, so an id line could bind to the PRECEDING block. The reset is now keyed to ANY heading level (anyHeadingRE), matching the item parser's own greedy-bind guard — the two components had disagreed about what closes an id-scan window. Honest boundary (§11.4.6): measured by an awk walk of both trackers comparing the two reset rules, **ZERO** id lines change attribution. This closes a divergence and a live trap for the moment the missing id lines are backfilled; it does not repair a mis-attribution happening today.
- **The ### A52 block code was declared by BOTH TMX-062 and TMX-071** — a genuine §11.4.19 one-item-one-block violation with measured data loss. Both blocks live in Fixed.md: the space-form

A52 NO-SUDO-... at line 2890 (TMX-062, itself a live specimen of the form this release made parseable) and the period-form ### A52. META-TEST-72-73-REGISTER-001 at line 3027 (TMX-071). The literal ### A52. occurs exactly ONCE; it is the CODE that repeats, across the two forms — measured at HEAD 5191e82, BEFORE the repair: `grep -c '^### A52\.' = 1`, `grep -cE '^### A52(\. | )' = 2`. Resolved by renumbering the TMX-071 block to ### A56. (A56 was free; the highest in use was A55), with the sync's ExplicitATM path RECORDING the identity rebind. Post-renumber those commands measure 0 and 1, and the owner map holds 25 keys. The superseded measurements are PINNED to their revision in the closure evidence rather than deleted, because the collision they document was real (§11.4.6).

- **TMX-078's stored identity named a block owned by a different ticket** — and no check could see it (landed in ec25327). The row stored {category:A, code_ordinal:50, location:Fixed} while Fixed.md:2945 ### A50 declared TMX-057, and TMX-078's real block was Issues.md:271 ### G5 SANITIZE-NAME-001. validate reported 0 findings throughout because no check compared a row's STORED identity against the MARKDOWN block it names — a missing check and a clean corpus return the identical quiet zero. It is not catchable DB-internally either: a duplicate-triple scan misses it, because TMX-078's triple IS unique (the colliding item carries the ordinal-0 sentinel the narrow parser left behind). Only the markdown can settle who owns a block. Repaired (one row; §9.2 pre-op backup before the UPDATE; DB WAL-checkpointed TRUNCATE before staging per §11.4.95) and covered by the new cross-surface detector ValidateBlockIdentity, wired into runValidate behind --issues / --fixed, with four false-positive guards each measured against the live corpus.
- **Closing an item could move its block identity onto a block another item already held — close.go wrote the destination regardless.** A close is a BLOCK-IDENTITY MIGRATION, not only a status change: the identity triple (current_location, category, code_ordinal) is a pointer to ONE markdown block, and flipping current_location re-points it from (Issues, cat, ord) to (Fixed, cat, ord). Block codes are CATEGORY-LOCAL and are NOT unique across the two trackers, so that destination can already be occupied — producing exactly the two-items-one-block state. ValidateBlockIdentity exists to report. CloseItem now checks the destination BEFORE any write, so the refusal is side-effect free and the row is left exactly as it was (§11.4.252 fail-closed), and the error names the colliding holder plus the concrete recovery (renumber to

a free code in `Issues.md`, re-sync, close again). Three false-positive guards mirror `validate_identity.go`'s, so the two components agree on what "claims a block" means (§11.4.201(1)): `code_ordinal <= 0` is the unknown-ordinal sentinel every add row carries and can never collide; an obsolete closure legitimately SHARES its successor's block (§11.4.90 — measured live, (Fixed, B, 3) is held by both TMX-001 and TMX-054); and the item's own row is excluded, so an idempotent double-close is never refused by itself. The migration is also RECORDED on the closure history event (block identity <CODE> migrated <from> -> Fixed, appended to any operator-supplied reason rather than replacing it) — a rewrite with no recorded evidence is indistinguishable from a row that was always that way (§11.4.226).

Changed

- **Two shipped governance gates that enforced NOTHING are now wired into the release gate.** `scripts/testing/tracker_structural_integrity.sh` and `scripts/review/review_round_record_test.sh` live outside `run_all.sh`'s `scripts/tests/[0-9][0-9]*.sh` glob, so `grep` proved **zero** invocations from any runner: both self-tested green by hand while gating zero (§11.4.205; §11.4.227(A) — a named gate must be implemented or a registered deferral, never silent debt). Found as a BLOCKING review finding and confirmed by re-measurement before acceptance. Both now execute from `scripts/verify.sh` as Layer-1 static gates CM-TRACKER-STRUCTURAL-INTEGRITY and CM-REVIEW-ROUND-RECORD-WIRED, each validated in BOTH polarities (golden-TRUE PASS on the healthy tree with resolved evidence; golden-FALSE FAIL with `rc=1` when the script is absent), each FAIL printing an actionable reproduce: line, and a FAIL of either `ABORTS.setup.sh` before the runtime suite (the shared `TRACKGOV_FAIL` block, which now also carries the third gate below).
- **The three code-less ### headings are REPAIRED, and a THIRD Layer-1 gate now covers the class so it cannot silently return.** The tracker grammar never required a `###` heading to carry a block code, so hand-added headings were absorbed into the preceding block's `raw_body` and were invisible to the §11.4.93 SSoT — one architectural cause reached by three unrelated triggers (a sub-note, the ATM→TMX prefix migration 89324dc, a §12.10 status log). All three now carry a parser-readable block code and a `**TMX-ID:**` line: `Issues.md`'s M24-ESCAPE-001 became `### B54 M24-`

ESCAPE-001 ... — FIXED (TMX-098), and Fixed.md's two became ### D2. (TMX-096) and ### D3. (TMX-097) — measured with `grep -c 'TMX-096\|TMX-097\|TMX-098' Issues.md Fixed.md` → Issues.md:0 Fixed.md:8, all three now resident in the closed tracker. The class is enforced by `scripts/testing/heading_grammar_gate.sh`, wired into `scripts/verify.sh` as Layer-1 static gate CM-HEADING-GRAMMAR (definition `scripts/verify.sh:935`, invocation `:1050`, its FAIL reaching `exit 1` through the same `TRACKGOV_FAIL` block as the two gates above). Measured live: Issues.md headings=12 coded=12 code-less=0; Fixed.md headings=86 coded=86 code-less=0; rc=0. It carries a §11.4.201(7)(b) control needle — 3 synthetic headings through the SAME scanner return seen=3 coded=2 code-less=1, so its zero is proven not blind — and a §11.4.115(F) RED/GREEN pair: against `HEAD:Issues.md + HEAD:Fixed.md` (the pre-repair artifact) it FAILs naming all three offenders by file and line, and against the repaired corpus it PASSes. Its self-test `scripts/testing/heading_grammar_gate_test.sh` reports `PASS=9 FAIL=0`, two of the nine being §1.1 mutations (classifier weakened → gate MISSES the code-less heading, rc=0; counter neutered → same), proving the verdict depends on the assertion and the count rather than on incidental output. Companion guide: `docs/scripts/heading_grammar_gate.md` (§11.4.18).

- **Tracker corpus repair** (operator-approved 2026-09-01, both decisions recorded): removed four stale ### G1..G4 duplicates present as full blocks in BOTH trackers (TMX-072..075, already closed in Fixed.md); resolved the A50 double-claim by renumbering Fixed.md's block to ### A55. with `**TMX-ID:**` TMX-078, then retiring the duplicate Issues.md ### G5. The emptied ## G. section was **ANNOTATED, not deleted**. Cross-tracker duplicates 5 → 0, enforced at the sync seam.
- **constitution submodule pointer b9096ac → f5876a3** — a codegraph ledger append; no governance text changed.
- **README.md** gains a doc-link row for the two newly-wired gates and their companion guides, so both are reachable from the canonical entry point (§11.4.212).

Tests

- **New Go test files:** `heading_forms_test.go` (the specified-oracle accept/refuse table plus a `headingRE/blockCodeRE` agreement test), `duplicate_block_test.go`, `upsert_identity_test.go`, `guard_coverage_test.go`, `validate_identity_test.go`,

validate_identity_duplicate_test.go,
add_hash_convention_test.go, add_category_validation_test.go,
db_to_md_add_rows_test.go, sourceless_signals_test.go; extended
parser_greedybind_test.go and reconcile_identity_test.go.

- **Test 51 gate hardening.** T3 asserted only that the output contains 0 findings — but a SKIPPED audit prints 0 findings too, so the gate would have PASSED on a check that never ran (§11.4.201 false-null). T3 now refuses skip-shaped output explicitly, and was proven load-bearing by mutation: hide the tracker → FAIL; restore → no false fire.
- **Mutation ledger (§1.1): 19 guards pinned**, each mutation verified to LAND in the file before being run; nine of the 19 were authored by the independent reviewer rather than by the author. Recorded in docs/qa/2026-09-01-heading-regex-widening/closure-evidence.md.
- **Blind tests caught by RUNNING the mutations rather than trusting the green.** Two of the author's own: the rebind test's assertions sat behind an inverted condition that skipped them every run; the precedence test asserted an outcome identical under both orderings. Two of ec25327's controls, for the same reason: a helper left HeadingHash empty so every fixture put () overwrote the previous row (the 2-item obsolete fixture collapsed to 1), and the ordinal-0 control asserted only findings==0, which the detector satisfies either way. All four were rewritten before being trusted.
- **Control needle (§11.4.201(7)(b)):** re-injecting the exact original TMX-078 collision into a DB copy makes the validator FAIL naming TMX-078, while the tracked DB reports 0 findings across 3 iterations — the live-corpus zero is proven not blind.

Validation

Every figure below is **pinned to the revision it was measured at**. The full verify.sh sweep HAS now completed on the current tree — its result, its enumerated skips, and its honest boundary are recorded below the table.

Measurement	Revision	Result
go vet ./...	ec25327	clean
go test -count=1 ./...	ec25327	ok
Test 51	ec25327	PASS=5 FAIL=0 SKIP=0 on 3/3 iteration
	ec25327	

Measurement	Revision	Result
db-to-md round-trip byte-identity		3/3, hash=9d91e1a1e871150aec5fa8eebea9a0
Sentinel rows (code_ordinal <= 0)	8dad4e3	17 → 3
Sentinel rows (category Z)	8dad4e3	4 → 0
Cross-tracker duplicates	8dad4e3	5 → 0
Sync idempotency	8dad4e3	unchanged=92, zero rebinds
validate findings	8dad4e3	0
verify.sh sweep	1690789	exit 0, GREEN, every summary FAIL=0 (hardware/topology, pre-existing)
verify.sh on the committable tree	current (2c8187a + uncommitted)	COMPLETED 2026-09-02T06:09Z — PASS=23, FAIL=0, SKIP=11, GREEN (verify_final earlier 05:42Z sweep predates later tree and does NOT validate this tree)
CM-TRACKER-STRUCTURAL-INTEGRITY (inside that sweep)	current	PASS — 6 tracker-structural check(
CM-REVIEW-ROUND-RECORD-WIRED (inside that sweep)	current	PASS — PASS=23 FAIL=0
CM-HEADING-GRAMMAR (inside that sweep)	current	PASS — no code-less tracker heading
heading_grammar_gate.sh standalone	current	PASS=2 FAIL=0 SKIP=0, rc=0; Issues.md 86/86 coded, Fixed.md 86/86 coded
heading_grammar_gate_test.sh (self-test)	current	PASS=9 FAIL=0 — incl. 2 §1.1 mutation RED on HEAD: artifacts
Full meta-test mutation sweep	current	NOT RUN against this tree
§11.4.185 manual QA	current	NOT PERFORMED

The completed sweep, in full (§11.4.6 honest boundary). It closed GREEN: tmux binary verified – safe to PATH-export. on HEAD 2c8187a plus the uncommitted engine / gate / corpus work described above. The **11 SKIPS are honest topology / capability skips (§11.4.3), not passes**, and are enumerated rather than aggregated:

- 08_oom_score_adj, 12_memory_pressure_under_cap, 13_tasksmx_stress, 14_concurrent_oom_independence — destructive; require TMX_TEST_DESTRUCTIVE=1 on a dedicated test host.

- 44_clipboard_copy_out_physical, 45_multiline_copy_physical, 46_paste_in_physical, 48_modifier_drag_override — require a GUI clipboard, absent on this headless host.
- 22_codegraph_mcp_wired, 32_ssh_dispatch_remote_nezha, 62_distribution_orchestrator — absent topology.

The sweep is a verdict on the automated suite for this tree and nothing more: it does NOT discharge §11.4.185 manual QA (still owed), it does NOT cover the 11 skipped topologies, and it does NOT stand in for the on-device / dual-host suites those skips name. **COMPLETED**

2026-09-02T06:09Z (verify_final.log, 1177 lines). An EARLIER sweep completed 05:42Z (verify_full.log); the tree was then edited further (.gitignore, README.md, CONTINUATION.md, Issues.md, Fixed.md, 14 regenerated exports), so the 05:42Z run does NOT validate this tree and the 06:09Z run does. verify.sh does read README.md and CONTINUATION.md, which is why the re-run was required rather than optional. Every edit after 06:09Z is documentation-only and is re-validated by the §11.4.40 full retest that gates the tag. `grep -rln 'CHANGELOG\|docs/changelogs' scripts/` matches two comment lines only (59_oomd_preference_avoid.sh:64, 61_no_libtinfo_version_warning.sh:45).

The 3 surviving code_ordinal <= 0 rows are **not** sentinels: they are TMX-038 / TMX-039 / TMX-041, the genuine ### A0. / ### B0 / ### C0 blocks at Fixed.md:2585 / 2652 / 2718, whose ordinal legitimately IS zero. The 14 rows that healed (TMX-057..066, TMX-072..075) were the true sentinels; the 14 newly-tracked blocks (TMX-079..092) are a DIFFERENT set of 14.

Accounting for 94 DB rows vs 92 parsed blocks (reproducible)

94 rows = 91 distinct block identities (location, CAT, ORD) + 2 rows sharing a block CODE with another row (TMX-001/TMX-054 at Fixed B3; TMX-071/TMX-062 at Fixed A52) + 1 row whose claimed identity has no block at all (TMX-050 at Fixed F1). 92 parsed = 91 distinct + 1 duplicated block CODE. All three extra rows are obsolete, exactly what the audit's Obsolete guard skips, which is why validate correctly reports 0 findings.

Instruments discarded en route (§11.4.201)

Three measurement attempts produced confident wrong numbers **without crashing**, and each is recorded because the failure shape is the lesson.

Reported	Why it was wrong
"62 rows missing from markdown"	joined rows to blocks on the TMX-ID literal, but 60 of 85 <code>Fixed.md</code> blocks declare no such line — it measured the known missing-id gap, not row absence (§11.4.201(9): an id literal read as a row key)
"79 rows unbound by heading hash"	reimplemented <code>computeHeadingHash</code> over the RAW heading remainder; <code>parser.go:201</code> hashes <code>cleanTitle</code> , with the trailing STATUS hint already stripped by <code>trailingStatusRE</code> . With the strip, <code>unbound = 2</code>
"the row gap is the two refused headings"	no DB row corresponds to either heading, so they contribute ZERO rows — the same §11.4.201(9) field-identity class as the first row

The first two were caught and discarded during the investigation. The third was **published** and is RETRACTED in 2c8187a and in the closure evidence: its conclusions (gap = 2, no live collision, `validate 0` findings) survive, but they were reached via the wrong rows, and the arithmetic above is the reproducible replacement.

Honest regression in one number

The identity-audit **unlocated count ROSE 55 → 57**. This is explained, not hidden: 14 newly-visible blocks entered the SSoT, and the audit can only LOCATE a block that declares a TMX-ID line, which 60 of 85 `Fixed.md` blocks (and 3 of the 8 `Issues.md` blocks censused at the time) do not. Those blocks assert no ownership and are skipped by design — counted and reported out loud, never silently zeroed (§11.4.201(6)).

Review trail (§11.4.134 iterate-to-GO)

The heading-regex batch went round 1 NO-GO (1 BLOCKING — the author's own corpus-removal loop over-deleted the `## H.` header and its §11.4.114 preamble; restored verbatim from `git show HEAD:Issues.md`) → round 2 NO-GO (3 IMPORTANT

- 3 MINOR, and it VERIFIED the round-1 BLOCKING closed) → round 3 NO-GO (1 IMPORTANT: the `itemContentEqual` identity gap, found in the seam the author had asked the reviewer to attack) → round 4 **GO** → delta review of the two filed items **GO** → delta review of the tracker/evidence text NO-GO on 3 MINOR documentation-accuracy findings, all three remediated in 1690789 and 2c8187a.

Round-numbering honest boundary (§11.4.6): no per-round artifact was written to disk, so the boundary could not be reconstructed from local state. The numbering is the reviewer's record, corroborated decisively by its own argument that a mutation which BECAME a round-2 finding cannot have survived round 1.

The review of the CURRENT batch is a separate, still-open loop — see below.

Known-open — explicitly NOT claimed fixed (§11.4.6)

1. **The independent review of the current batch returned NO-GO** — the LATEST round (2026-09-02, §11.4.209 substrate) reported 0 BLOCKING, 5 IMPORTANT, 1 MINOR (six findings total — F1 through F6 in CONTINUATION.md §3.37b; corrected 2026-09-29, this document previously undercounted "4 IMPORTANT" here, one short of the correct, enumerated count that CONTINUATION.md and CHANGELOG.md both already carried correctly), all documentation/hygiene, code layer clean; the PRIOR round reported 3 BLOCKING, 6 IMPORTANT, 5 MINOR — and has NOT returned a clean GO. §11.4.134 requires iterating to ZERO findings AND ZERO warnings. All three BLOCKING findings were confirmed by independent re-measurement before acceptance and are fixed in this release (the two unwired gates; a CONTINUATION.md freshness stamp bumped with zero content change; a closure-evidence citation this same batch falsified).
2. **The four reviewer-authored mutations that SURVIVED the prior round are REMEDIATED** (`cmd/workable-items/`

sourceless_signals_test.go). The LATEST round ran 9 reviewer-authored mutations and measured 8 CAUGHT, 1 SURVIVED — and that survivor is the pre-declared, re-measured honest boundary the author documented in that same file (the owners[it.ATMID] half of blockClaimedBy is defensive redundancy, not a load-bearing signal), NOT an escape. The §11.4.194(6)(d) pattern that produced the prior round's four survivors is recorded as the reason reviewer-authored mutations are mandatory, not as a closed-by-assertion item.

3. **close.go's collision guard REFUSES a bad closure; it does not RENUMBER one.** The guard stops a NEW two-items-one-block state from being written and names the concrete recovery, but the renumber is manual — an automatic free-code allocation on close is not implemented and is not claimed. Nor does it repair an EXISTING stale pointer: TMX-050 still names F1, a block existing in NEITHER tracker. That repair is OWED.
4. **The missing TMX-ID backfill is OWED**, which is why **57 of 95 items remain unlocatable by the identity audit**. That number is reported out loud by `validate` rather than counted clean, and it is not claimed to be shrinking this cycle.
5. **The full meta-test mutation sweep has NOT been run against this tree.** Its outcome is not known and is not claimed. The 19 individually-pinned §1.1 mutations recorded in the closure evidence are not a substitute for the whole-suite mutation sweep. *(Two items formerly listed here are now CLOSED and moved to "Changed" / "Validation": the three code-less ### headings are repaired and the class is covered by the wired CM-HEADING-GRAMMAR gate; and the full `verify.sh` sweep on this tree COMPLETED GREEN — PASS=79 FAIL=0 SKIP=11.)*
6. **§11.4.185 manual QA has NOT been performed**, and the tag has NOT been cut. Both remain preconditions of the release, alongside the §11.4.40 full retest.
7. **This cycle closes NONE of the v1.0.44 known-open items.** Test 68 C6/C7 (TMX-080 / TMX-081 family), TMX-090 (17_scrollback_copy_mode.sh T3 live WheelUpPane binding), test 71 C4 on the 3.7b artifact, and the Fixed.md \$L placeholder closure SHAs all remain open exactly as recorded in the v1.0.44 release entry.
8. **§11.4.44 revision headers are absent from Issues.md, Fixed.md and CONTINUATION.md** (TMX-101, Queued). Measured 2026-09-02: Issues.md 0/0, Fixed.md 0/0, CONTINUATION.md 0 Revision / 1 Last-updated; control needle docs/workable-items/Status.md = 1, so the detecting grep is not blind. Repo-wide 60 of 86 tracked .md carry the field. PRE-EXISTING — not introduced by this batch, and NOT fixed in it.

9. **Issues_Summary.md and Fixed_Summary.md do not exist anywhere in the repository** (TMX-102, Queued), though §11.4.12 and §11.4.53 mandate both and scripts/tests/51_workable_items_db_integrity.sh references them. `git ls-files | grep -iE 'Issues_Summary|Fixed_Summary'` is empty while the same search finds `Issues.md` and `Fixed.md` (control needle). PRE-EXISTING — not introduced by this batch, and NOT fixed in it.
10. **sync db-to-md WRITES the git-tracked SSoT DB on every run**, including a pure verification round-trip that renders into a throwaway `--out-dir` (TMX-103, Queued). `cmd/workable-items/sync_db_to_md.go:131-134` rewrites `meta.last_sync_direction / meta.last_sync_timestamp` unconditionally, so the documented byte-identity verification procedure dirties the tracked `docs/workable_items.db` and cannot be restored byte-exactly. Scope proof: successive runs on copies differ ONLY in those two meta rows (full logical table dump); `validate` is byte-neutral. PRE-EXISTING at HEAD: `cmd/workable-items/sync_db_to_md.go` — this batch newly enshrines the round-trip as the verification mechanism, and does NOT fix the side effect.
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Discovery direction (§11.4.238)

None of this cycle's defects was operator-reported. The heading-form gap surfaced from a tracker-integrity investigation; six more came out of fixing it; TMX-093 / TMX-094 / TMX-095 and the two audit defects came from a five-stream root-cause fan-out (§11.4.230(C)); and the three BLOCKING findings — including two gates that enforced nothing — came from the independent review. That is the intended direction of discovery, and it is also why the review loop is not yet closed.

Files changed

- **Engine:** `cmd/workable-items/{parser.go, reconcile.go, db.go, sync_md_to_db.go, sync_db_to_md.go, add.go, close.go, validate_identity.go, main.go}`.
- **Tests:** `cmd/workable-items/{heading_forms, duplicate_block, upsert_identity, guard_coverage, validate_identity, validate_identity_duplicate, add_hash_convention, add_category_validation, db_to_md_add_rows, sourceless_signals, close_identity, parser_greedybind, reconcile_identity}_test.go`; `scripts/tests/51_workable_items_db_integrity.sh`.

- **Gates:** scripts/verify.sh; scripts/testing/tracker_structural_integrity{,_test}.sh; scripts/review/review_round_record{,_test}.sh; scripts/testing/heading_grammar_gate{,_test}.sh.
- **Corpus + SSoT:** Issues.md, Fixed.md, docs/workable_items.db.
- **Docs:** CHANGELOG.md, CONTINUATION.md, README.md, docs/scripts/{tracker_structural_integrity,review_round_record,heading_grammar_gate}.md, docs/qa/2026-09-01-heading-regex-widening/closure-evidence.md, this file.
- **Submodule:** constitution pointer b9096ac → f5876a3.
- **Pending at the time of writing:** VERSION (1.0.45 / versionCode 46) is bumped when the tag is cut, not here.

Sources verified 2026-09-01

NO external solution found — original work (§11.4.8). Every defect above is internal to this project's own tracker engine, gate wiring, and corpus; no third-party documentation, API, or upstream behaviour was consulted or relied on, so there is no external source to cross-reference under §11.4.99.